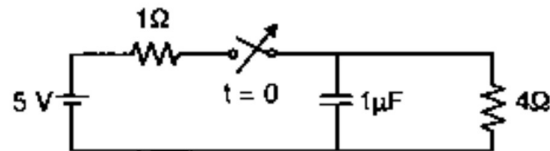


Question 1

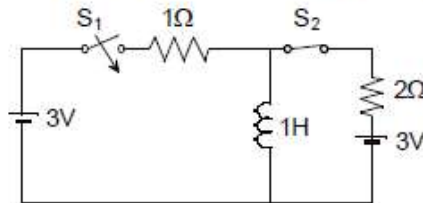
The switch in the circuit has been closed for a long time. It is opened at $t = 0$. At $t = 0^+$, the current through the $1\ \mu\text{F}$ capacitor is



Also find the current through capacitor at $t = 4\ \mu\text{s}$ and $t = 8\ \mu\text{s}$

Question 2

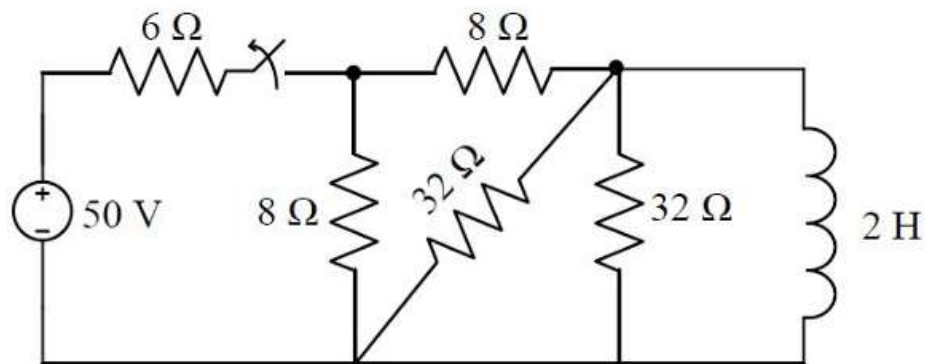
In the circuit shown, switch S_2 has been closed for a long time. At time $t = 0$ switch S_1 is closed.



Determine the value of current through inductor at $t = 0^+$ and $t = \infty$. Determine the expression for the current through the inductor for $t > 0$

Question 3

The switch in the figure below was closed for a long time. It is opened at $t = 0$. The current in the inductor of $2\ \text{H}$ for $t \geq 0$, is



(A) $2.5e^{-4t}$

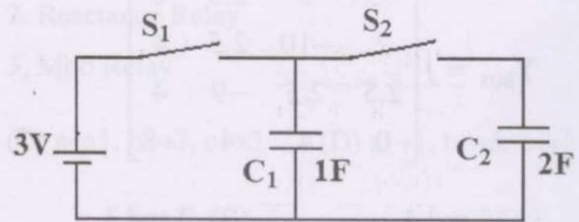
(B) $5e^{-4t}$

(C) $2.5e^{-0.25t}$

(D) $5e^{-0.25t}$

Question 4

In the figure shown, all elements used are ideal. For time $t < 0$, S_1 remained closed and S_2 open. At $t = 0$, S_1 is opened and S_2 is closed. If the voltage V_{C_2} across the capacitor C_2 at $t = 0$ is zero, the voltage across the capacitor combination at $t = 0^+$ will be



Clue: The total energy stored in the capacitors remains same.

Challenge question

Question 5

- **25** For the circuit shown in Fig. 6-28, let $v_s = 4$ V and find: (a) $v_1(t)$; (b) $v_2(t)$; (c) $i_3(t)$.

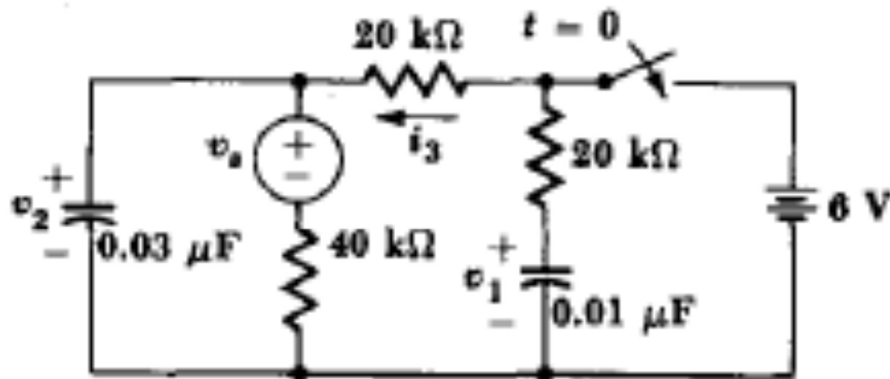


Fig. 6-28 See Probs. 24 and 25.

The circuit which is active for $t > 0$ is (a) two separate 1st order circuits, (b) one second order circuit. (Hint: To find that, zero all forcing functions and draw the resultant circuit)

If there are two 1st order circuits independently solve them

Answers

- 1) 1.25 A, 0.459 A, 0.169 A
- 2) 1.5 A, 4.5 A, and $4.5 - 3 e^{-(t/1.5)} \text{ A}$
- 3) C
- 4) 1.732 V
- 5) Two separate 1st order RC circuits

$$\begin{aligned}
 v_1(0^+) &= 4 \text{ V} & v_1(\infty) &= 5.33 \text{ V} & v_1 &= 5.33 - 1.33 e^{-t/200 \mu\text{s}} \\
 v_2(0^+) &= 4 \text{ V} & v_2(\infty) &= 6 \text{ V} & v_2 &= 6 - 2 e^{-t/400 \mu\text{s}} \\
 i_3(0^+) &= 100 \mu\text{A} & i_3(\infty) &= 33.33 \mu\text{A} & i_3 &= 33.33 + 66.67 e^{-t/400 \mu\text{s}} \\
 \tau_1 &= (0.01 \mu\text{F}) \times 20 \text{ k}\Omega = 200 \mu\text{s} & \tau_2 &= 0.03 \mu\text{F} \times [20 \parallel 40] \text{ k}\Omega \\
 & & & & &= 400 \mu\text{s}
 \end{aligned}$$